# Pinned group summary: `SL(n=2)`

Pinning-Sympy automated output

July 6, 2026

## Contents

# 1 Basic attributes

| | |
|---|---|
| Group | `SL(n=2)` |
| Matrix size | $2 \times 2$ |
| Rank | 1 |

## 1.1 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} \\ x_{10} & -x_{00} \end{bmatrix}$$

## 1.2 Generic torus element

$$\begin{bmatrix} t_0 & 0 \\ 0 & \frac{1}{t_0} \end{bmatrix}$$

## 1.3 Trivial characters

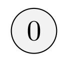
$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

# 2 Root system

### Summary Properties

| | |
|---|---|
| Dynkin type | `A1` |
| Reduced | True |
| Simply laced | True |
| Number of roots | 2 |

## 2.1 Dynkin diagram




## 2.2 Cartan matrix

$$\begin{bmatrix} 2 \end{bmatrix}$$

## 2.3 Roots and coroots

| **Root** | **Coroot** | **Simple** | **Sign** | **Multipliable** |
|---|---|---|---|---|
| $(-1,\ 1)$ | $(-1,\ 1)$ | No | $-$ | No |
| $(1,\ -1)$ | $(1,\ -1)$ | Yes | $+$ | No |

## 2.4 Linear combinations of roots

**Definition 2.1.** For two roots $\alpha, \beta$ in a root system $\Phi$, the set of positive integral linear combinations of $\alpha, \beta$ that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

No pairs of roots generate positive integer linear combinations.